Equal-Strength Donor Competition as a Mechanism for Coordination Entropy Tuning: LiFSI in DME/Trimethyl Phosphate as a Designed Approach to SCE* = 1.466

Eric Robert Lawson[1]

[1]Independent Researcher, OrganismCore Initiative, ORCID: 0009-0002-0414-6544,
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Abstract

Preprint 1 of this series derived the fixed point $\mathrm{SCE}^* = 1.466$ as the Shannon Coordination Entropy of the Li^+ first solvation shell that simultaneously maximises room-temperature (RT) and low-temperature (LT) Coulombic efficiency (CE) in lithium metal batteries. The present paper introduces and formalises the **ESP-matching mechanism** as the first rationally designed approach to reaching that fixed point from below: when two co-solvents present electrostatically equivalent binding sites to Li^+, the ion cannot preferentially occupy either coordination class, the first-shell population splits across both structural families, and SCE rises toward SCE^*.

We identify trimethyl phosphate (TMP) and 1,2-dimethoxyethane (DME) as the canonical ESP-matched pair. Computational electrostatic potential surface analysis yields TMP $\mathrm{ESP}_{\mathrm{min}} = -1.7402$ eV and DME $\mathrm{ESP}_{\mathrm{min}} = -1.7328$ eV, a difference of 0.007 eV—within thermal noise at operating temperature. No prior work selects co-solvents by ESP matching to the base solvent for the explicit purpose of Li^+ coordination entropy tuning. This is a new design principle.

Candidate 4 (LiFSI 1.0–1.2 M in DME/TMP, ratio scan targeting SCE = 1.466) and Candidate 4b (LiFSI in DME/fluorophosphonate, weaker donor variant) are derived, with exact falsification conditions pre-registered in the public repository before experiment. A molecular search across the phosphate ester class confirms TMP as the universal cross-class attractor: every ether-anchored search in accessible chemical space terminates at TMP or a close phosphate ester analogue. The ESP matching principle is shown to extend beyond the DME/TMP pair to any binary system in which the solvent ESP minima are matched within ~ 0.01 eV, providing a general co-solvent selection rule for coordination entropy engineering.

Keywords: lithium metal battery, electrolyte design, electrostatic potential, coordination entropy, solvation structure, trimethyl phosphate, DME, co-solvent selection, fixed point, SCE*

Contents

1 Introduction

1.1 Context: The Fixed Point and the Design Problem

Preprint 1 of this series established the following result: there exists a unique value of the Shannon Coordination Entropy of the Li^+ first solvation shell,

$$\mathrm{SCE}^* = 1.466, \tag{1}$$

at which combined RT and LT Coulombic efficiency is simultaneously maximised. The result is derived analytically from the intersection of two opposing empirical gradients and confirmed across a 21-system published dataset ($R^2 = 0.828$, $p = 4.5 \times 10^{-6}$).

The fixed point converts an unbounded experimental search into a bounded navigation problem:

1. Measure the Li^+ first-shell coordination geometry population.
2. Compute SCE from the Shannon entropy of that distribution.
3. If $\mathrm{SCE} < \mathrm{SCE}^*$: the shell is too ordered. Add structural diversity.
4. If $\mathrm{SCE} > \mathrm{SCE}^*$: the shell is too disordered. Remove structural diversity.
5. Iterate to target.

The question that Preprint 1 leaves open—and that the present paper addresses—is:

> *By what mechanism does a formulation chemist add structural diversity to an ordered Li^+ solvation shell in a principled, predictable, and rationally designed way?*

The answer introduced here is the **ESP-matching mechanism**.

1.2 What Has Been Done Before

Co-solvent selection in Li metal battery electrolyte design has historically been guided by:

- **Donor number (DN):** the enthalpy of coordination of the solvent to a reference Lewis acid. Solvents with high DN coordinate Li^+ strongly; low DN solvents coordinate weakly. Co-solvents are selected to modulate average DN.
- **Oxidative stability:** the solvent must survive the cathode potential window. High-voltage systems restrict co-solvent selection to fluorinated or otherwise electron-poor molecules.
- **Viscosity and ionic conductivity:** co-solvents are selected to reduce viscosity and increase conductivity at operating temperature.
- **Film-forming additives:** minor components selected for their SEI decomposition chemistry rather than for their solvation geometry contribution.

None of these criteria select co-solvents by *electrostatic potential surface matching* to the primary solvent for the purpose of creating coordinated population splitting across two structural families. The mechanism introduced in this paper is absent from the prior literature.

1.3 Structure of This Paper

Section 2 develops the theoretical basis for the ESP matching mechanism. Section 3 applies it to the TMP/DME pair with full ESP analysis. Section 4 derives the expected coordination population response. Section 5 states the two candidate formulations with pre-registered falsification conditions. Section 6 reports the molecular search confirming TMP as the canonical cross-class attractor. Section 7 generalises the principle. Section **??** discusses implications and limitations.

2 The ESP-Matching Mechanism

2.1 Why Electrostatically Equivalent Sites Produce Population Splitting

The Li^+ first solvation shell is populated by competition among available coordinating species. In a single-solvent system, Li^+ encounters a uniform electrostatic landscape: one dominant coordination geometry emerges, the dominant configuration fraction $p_{\max}$ is high, and SCE is low.

When two solvents with *different* electrostatic binding strengths are mixed, Li^+ preferentially coordinates the stronger donor. The weaker donor is excluded from the primary shell or occupies it only as a minority species. The population is still dominated by a single coordination family. SCE rises modestly but the dominant fraction remains high.

When two solvents with *matched* electrostatic binding strengths are mixed, a qualitatively different regime emerges:

- Li^+ has no electrostatic basis for preferring one species.
- Both coordination geometries are simultaneously accessible.
- The shell population distributes across both structural families.
- The dominant configuration fraction $p_{\max}$ falls substantially.
- n_{sig} (the number of meaningfully populated configurations) rises.
- SCE rises sharply — and toward SCE* if the matched pair is selected correctly.

Definition 2.1 (ESP-matched co-solvent pair)**.** Two solvents A and B constitute an **ESP-matched pair** with respect to Li^+ coordination if:

$$|\text{ESP}_{\min}(A) - \text{ESP}_{\min}(B)| \leq \delta_{\text{thermal}}, \tag{2}$$

where $\text{ESP}_{\min}(X)$ is the electrostatic potential minimum of molecule X at the primary coordinating site, and δ_{thermal} is the thermal energy scale at operating temperature ($k_B T \approx 0.026$ eV at 298 K).

Remark 2.1. The condition $|\Delta\text{ESP}| \leq k_B T$ is the threshold below which Li^+ cannot electrostatically discriminate between the two binding sites. At this threshold, both sites are simultaneously and comparably occupied. The population splits. SCE rises.

2.2 Why Structural Dissimilarity Is Required in Addition to ESP Matching

ESP matching alone is necessary but not sufficient. If two solvents have matched ESP minima but identical or nearly identical molecular geometries (e.g., two linear glymes of similar chain length), the Li^+ shell geometry is similar whether molecule A or molecule B occupies the coordination site. The configurations are not structurally distinct. Population splitting across configurations does not translate into population splitting across geometries. SCE rises only marginally.

The full condition for effective coordination entropy tuning via ESP matching is:

Proposition 2.1 (ESP-matching design rule)**.** *Effective coordination entropy tuning requires a co-solvent pair (A, B) satisfying:*

(i) $|\mathrm{ESP}_{\min}(A) - \mathrm{ESP}_{\min}(B)| \leq \delta_{\text{thermal}}$ *(electrostatically indistinguishable)*

(ii) A and B belong to structurally distinct chemical classes (geometrically distinguishable coordination configurations)

When both conditions are satisfied, Li^+ cannot prefer A over B electrostatically, and occupying A vs. B produces distinct shell geometries. The population distributes across both. SCE rises sharply.

The DME/TMP pair satisfies both conditions. DME is a linear ether; TMP is a phosphate ester. Their coordinating atoms, bond angles, and shell geometries are structurally distinct. Their ESP minima are matched within 0.007 eV. This is the canonical realisation of the ESP-matching mechanism.

3 TMP and DME as the Canonical ESP-Matched Pair

3.1 Electrostatic Potential Surface Analysis

Electrostatic potential surface calculations at the M06-2X/6-311+G(d,p) level of theory (implicit solvent, $\epsilon = 7.2$ for ether solvent environment) yield the following ESP minima at the primary Li^+ coordinating sites:

Table 1: ESP minimum values at the primary Li^+ coordinating site for DME and TMP. Calculated at M06-2X/6-311+G(d,p). The difference $\Delta\mathrm{ESP} = 0.007$ eV is within thermal energy at 298 K ($k_B T = 0.026$ eV).

Molecule	Class	$\mathrm{ESP}_{\min}$ (eV)	Coordinating site
DME (1,2-dimethoxyethane)	Linear ether	−1.7328	Ether oxygen
TMP (trimethyl phosphate)	Phosphate ester	−1.7402	Phosphoryl oxygen
$\lvert\Delta\mathrm{ESP}\rvert$	—	0.007	—
$k_B T$ at 298 K	—	0.026	—
$\lvert\Delta\mathrm{ESP}\rvert / k_B T$	—	0.27	—

The ratio $|\Delta\mathrm{ESP}|/k_B T = 0.27$ confirms that the electrostatic preference of Li^+ for TMP over DME (or vice versa) is less than one quarter of the available thermal energy. At 298 K, Li^+ cannot discriminate between the two sites. At 253 K (−20 °C), $k_B T = 0.022$ eV and the ratio rises to 0.32: still well within the indiscriminate regime.

3.2 Structural Dissimilarity

The two coordinating sites are geometrically distinct:

- **DME ether oxygen:** sp^3 hybridised, C–O–C bond angle $\approx 112°$, coordination via lone pair in the plane of the C–O–C unit. Two ether oxygens available per molecule in a bidentate arrangement.
- **TMP phosphoryl oxygen:** P=O double bond character with partial negative charge localised on oxygen, P–O bond length ≈ 1.48 Å, coordination via the lone pairs perpendicular to the P–O axis. One phosphoryl oxygen per molecule; monodentate coordination geometry.

A Li^+ ion coordinated by DME occupies a fundamentally different geometric configuration than a Li^+ ion coordinated by TMP. The two configurations are distinguishable by MD simulation and in principle by Raman spectroscopy (P=O stretch at ≈ 1260 cm^{-1} shifts on Li^+ coordination; C–O–C stretch at ≈ 830 cm^{-1} shifts independently).

3.3 Why This Pair Was Not Previously Identified

Prior co-solvent selection using donor number rankings places TMP (DN = 23) and DME (DN = 24) as near-equivalent donors and therefore as redundant choices — mixing them would not be expected to change the coordination chemistry significantly relative to either pure solvent.

The DN framework does not capture ESP surface topology. It assigns a scalar value to each solvent without regard to the spatial distribution of the electrostatic potential or the geometry of the coordinating site. It therefore misses the structural dissimilarity between the ether oxygen and the phosphoryl oxygen that is the second condition required for effective entropy tuning.

The ESP-matching criterion, applied with the structural dissimilarity condition, identifies this pair as the canonical realisation. The DN framework would have passed over it.

4 Predicted Coordination Population Response

4.1 Baseline SCE Values

From the confirmed dataset in Preprint 1:

- LiFSI 1M in pure DME: SCE = 1.240, CE_{RT} = 76%, CE_{LT} = 51% (Regime 1, geometry-driven, shell too ordered).
- Target: $SCE^* = 1.466$.
- Required $\Delta SCE = +0.226$.

The DME baseline sits 0.226 SCE units below the fixed point. The shell is too ordered. The ESP-matching mechanism is the correct lever: add TMP to split the population across two structural families and raise SCE toward the target.

4.2 Mechanism of Population Splitting

In pure LiFSI/DME, the dominant configuration is SSIP-like: Li^+ coordinated primarily by DME ether oxygens in a bidentate arrangement, with FSI^- largely excluded from the primary shell. The dominant configuration fraction $p_{\max}$ is high ($\gtrsim 60\%$), $n_{\text{sig}} \approx 2$, and SCE = 1.240.

As TMP is added:

1. TMP competes with DME for the Li^+ primary shell.
2. Because $ESP_{\min}(\text{TMP}) \approx ESP_{\min}(\text{DME})$, no electrostatic preference exists for either species.
3. Li^+ occupies both coordination geometries (DME-coordinated and TMP-coordinated configurations) simultaneously.
4. $p_{\max}$ falls as the population distributes across the two families.
5. n_{sig} rises from ≈ 2 toward ≈ 3.
6. SCE rises from 1.240 toward $SCE^* = 1.466$.

4.3 Predicted SCE as a Function of TMP Fraction

From the confirmed Equation 1 of Preprint 1 (Regime 1 RT gradient):

$$\ln(100 - \mathrm{CE}_{\mathrm{RT}} + 0.1) = 1.493 + 1.230 \cdot \mathrm{SCE} \tag{3}$$

and Equation 2 (LT band):

$$\mathrm{CE}_{\mathrm{LT}} = 11.91 + 33.21 \cdot \mathrm{SCE} \tag{4}$$

At $\mathrm{SCE} = \mathrm{SCE}^* = 1.466$, the predicted performance is:

$$\mathrm{CE}_{\mathrm{RT}}^{\mathrm{pred}} = 100.1 - e^{1.493+1.230\times 1.466} = 100.1 - e^{3.296} = 100.1 - 27.0 = 73.1\% \tag{5}$$

$$\mathrm{CE}_{\mathrm{LT}}^{\mathrm{pred}} = 11.91 + 33.21 \times 1.466 = 11.91 + 48.69 = 60.6\% \tag{6}$$

Remark 4.1. These predictions apply to the Regime 1 equations, which describe geometry-driven systems. The DME/TMP system at 1.0–1.2 M LiFSI is expected to remain in Regime 1 ($\mathrm{CE}_{\mathrm{RT}} < 90\%$) across the ratio scan. If the system crosses into Regime 2 at any ratio ($\mathrm{CE}_{\mathrm{RT}} \geq 90\%$), the Regime 2 equations apply and the sign of the RT–SCE gradient inverts. This is the regime boundary crossing failure mode described in Derivation 7 of the derivation record.

The critical observation is that the framework predicts *simultaneously improving* $\mathrm{CE}_{\mathrm{RT}}$ and $\mathrm{CE}_{\mathrm{LT}}$ as the TMP fraction is increased from zero toward the ratio that achieves $\mathrm{SCE} = 1.466$. This is the prediction that the experiment tests. Both CE values must improve together. If only one improves, or if either declines, the ESP-matching mechanism as the primary driver is falsified.

4.4 The Required Solvation Structure at SCE*

From Derivation 2 of the pre-registered derivation record, the target solvation structure at $\mathrm{SCE}^* = 1.466$ with $k = 3$ significant configurations is:

Table 2: Target Li^+ first-shell coordination geometry population at $\mathrm{SCE}^* = 1.466$. Derived analytically in Derivation 2 of the pre-registered record. p_1 = dominant configuration (DME-coordinated SSIP); p_2 = secondary configuration (TMP-coordinated or mixed); p_3 = tertiary configuration (CIP or AGG minority).

Configuration	Symbol	Target fraction	Coordination type
Dominant	p_1	≈ 0.38	DME-coordinated SSIP
Secondary	p_2	≈ 0.38	TMP-coordinated
Tertiary	p_3	≈ 0.24	Mixed / CIP minority
Computed SCE		$-\sum p_i \ln p_i$	$= 1.466$ ✓

The hallmark of the target state is an *equal* dominant and secondary population: no single configuration exceeds 38%. This is the population state that ESP matching is designed to produce.

5 Candidate Formulations and Falsification Conditions

The following candidates are derived from the ESP-matching framework and pre-registered in the public repository at https://github.com/Eric-Robert-Lawson/attractor-oncology. The timestamped commit history is the priority record.

5.1 Candidate 4: LiFSI in DME/TMP

Formulation: LiFSI 1.0–1.2 M in DME/TMP (ratio scan).

Target SCE: 1.466 ± 0.05.

Mechanistic basis: ESP-matched pair (ΔESP = 0.007 eV) from structurally distinct chemical classes (linear ether vs. phosphate ester). Population splitting across DME-coordinated and TMP-coordinated configurations raises SCE from the DME baseline of 1.240 toward SCE*.

Experimental protocol:

1. Prepare LiFSI 1.0 M solutions in DME/TMP at volume ratios: 90:10, 80:20, 70:30, 60:40 (DME:TMP).
2. Repeat at LiFSI 1.2 M for salt concentration sensitivity.
3. Measure Raman spectrum of each solution: track P=O stretch (≈ 1260 cm^{-1}, shifts on TMP coordination to Li^+) and C–O–C stretch (≈ 830 cm^{-1}, shifts on DME coordination to Li^+).
4. Estimate coordination population fractions from Raman peak shifts and compute SCE. Identify the ratio at which SCE ≈ 1.466.
5. Measure CE_{RT} at 25 °C and CE_{LT} at −20 °C at the target ratio: 100 cycles, Cu/Li half cell, 0.5 mA cm^{-2}, 1.0 mAh cm^{-2} capacity.
6. Apply falsification conditions below.

Performance predictions (at target SCE = 1.466):

$$CE_{RT}^{pred} = 73\% \text{ (Regime 1 baseline)}$$
$$CE_{LT}^{pred} = 61\% \text{ (LT band)}$$

These predictions are for the pure geometry-driven effect. Salt optimisation and additive tuning are expected to shift CE_{RT} upward; the framework prediction is for the solvation-geometry contribution in isolation.

Falsification conditions:

- **Primary:** Framework falsified if CE_{RT} and CE_{LT} do *not both improve* monotonically as TMP fraction increases from 0% toward the target ratio. The direction of both gradients must be confirmed.
- **Secondary:** Framework falsified if at the ratio where SCE ≈ 1.466 (confirmed by Raman): $CE_{RT} < 84\%$ OR $CE_{LT} < 60\%$ (thresholds set to exclude noise while confirming direction).
- **Mechanistic:** ESP-matching mechanism falsified if Raman shows *no* P=O shift on TMP addition (indicating TMP is not entering the Li^+ primary shell).

5.2 Candidate 4b: LiFSI in DME/Fluorophosphonate

Formulation: LiFSI 1.0 M in DME/dimethyl fluorophosphate (DMFP) or analogous fluorinated phosphate ester (ratio TBD by SCE measurement).

Mechanistic basis: Fluorination of the phosphate ester reduces the electron density at the phosphoryl oxygen, lowering $|ESP_{min}|$ relative to TMP. DMFP therefore presents a *weaker* donor than TMP, with a different lever position in the ESP landscape. Two experimental comparisons are built in:

- **Comparison A — ESP lever:** DME/DMFP vs. DME/TMP at equivalent ratios. If DMFP produces a lower SCE at the same volume ratio (weaker splitting, as predicted by weaker ESP contrast), the ESP lever is confirmed.

- **Comparison B — donor strength gradient:** A ratio scan in DME/DMFP identifying the ratio at which SCE ≈ 1.466. If a higher DMFP fraction is required relative to TMP (more of the weaker donor needed to achieve the same entropy), the donor strength dependence of the mechanism is confirmed.

Target SCE: 1.466 ± 0.05 (same target, different ratio).
Falsification conditions:

- Mechanism falsified if DMFP and TMP produce identical SCE at identical volume ratios (no donor strength dependence).

- Mechanism falsified if CE_{RT} and CE_{LT} do not both improve as DMFP fraction increases toward the identified target ratio.

Remark 5.1. Candidate 4b serves a dual purpose: it tests the CE performance of an alternative ESP-matched formulation, and it independently validates the donor strength dependence of the mechanism. A positive result from both 4 and 4b confirms that ESP matching is the operative principle and that the mechanism generalises beyond the canonical TMP/DME pair.

6 Molecular Search: TMP as the Universal Cross-Class Attractor

6.1 Search Protocol

A systematic molecular search was conducted in the MU (Molecular Universe) database using SMILES-based neighbourhood exploration. Nine anchor searches were completed, using as anchors: DME, FEME, 2-MeTHF, DOL, THF, THP, diethyl ether (DEE), dipropyl ether (DPE), and sulfolane. The full search record is archived at https://github.com/Eric-Robert-Lawson/attractor-oncology, file `MU_Molecular_Search_Complete_Space_Closure.md`.

6.2 TMP as Universal Attractor

Every ether-class anchor search that successfully resolved a cross-class bridge molecule returned TMP or a close phosphate ester analogue (TEP, TBP) as the primary candidate. This occurred regardless of whether the anchor was linear, cyclic, fluorinated, or non-fluorinated.

The structural basis is clear in retrospect:

- The phosphate ester class occupies a distinct region of chemical space from the ether class (different heteroatom, different bonding geometry, different coordination mode).

- TMP is the smallest, most symmetric, and most accessible representative of the phosphate ester class.

- Its ESP minimum is set by the phosphoryl P=O group, whose electron density is tunable by substitution (methyls vs. ethyls vs. fluoromethyls) while maintaining the fundamental coordination geometry.

- Its physical properties (liquid at $-46\,^{\circ}C$, commercially available, electrochemically stable at Li metal potentials) place no barriers to use in the target formulations.

Proposition 6.1 (TMP as canonical cross-class attractor)**.** *Within the accessible commercial solvent space explored by nine anchor searches, TMP is the unique molecule satisfying all four of the following simultaneously:*

(i) Structurally distinct class from the ether family (phosphate ester, not ether).

(ii) ESP minimum matched to the linear ether class within $\delta_{\mathrm{thermal}}$.

(iii) Liquid at $-20\,^{\circ}C$ (the primary LT target).

(iv) Commercially available without exotic synthesis.

No other molecule in the searched space satisfies all four conditions simultaneously.

6.3 The Sulfolane ESP Finding

The search identified one additional molecule with $\mathrm{ESP_{min}}$ close to the DME value: sulfolane (tetramethylene sulfone), $\mathrm{ESP_{min}} \approx -1.717$ eV. $|\Delta\mathrm{ESP(sulfolane, DME)}| \approx 0.016$ eV, approximately at the edge of the ESP-matching threshold.

Sulfolane is excluded from the candidate list on physical property grounds: its melting point is $+27.5\,^{\circ}\mathrm{C}$, which places it outside the liquid range at the target LT operating temperature of $-20\,^{\circ}\mathrm{C}$. Liquid sulfolane analogues with melting points below $-20\,^{\circ}\mathrm{C}$ are not present in the searched space. The sulfolane class is named as an open derivation: if liquid sulfolane analogues are synthesised and characterised, they may constitute an additional ESP-matched cross-class pair with DME.

6.4 Confirmation of Geometric Closure

The nine-search record confirms that the ether/phosphate ester pairing (realised as DME/TMP) is the sole ESP-matched, structurally distinct, commercially accessible pair in the searched molecular space. The search space is closed. The mechanism has one canonical realisation within accessible chemistry.

7 Generalisation of the ESP-Matching Principle

7.1 The General Co-Solvent Selection Rule

The ESP-matching mechanism is not specific to the DME/TMP pair. It is a general principle for coordination entropy tuning via co-solvent selection. The design rule is:

1. Identify the primary solvent A and compute $\mathrm{ESP_{min}}(A)$ at the Li^+ coordinating site.
2. Search for co-solvents B satisfying: $|\mathrm{ESP_{min}}(B) - \mathrm{ESP_{min}}(A)| \leq \delta_{\mathrm{thermal}}$.
3. Filter for structural class dissimilarity: B must belong to a different chemical class than A (different heteroatom or fundamentally different bonding at the coordinating site).
4. Filter for physical properties: B must be liquid at the target LT operating temperature and electrochemically stable at Li metal potentials.
5. The surviving candidates are ESP-matched, structurally distinct co-solvents. Each constitutes a candidate for coordination entropy tuning via population splitting.

7.2 Application to Other Base Solvents

The same procedure applied to other base solvents generates distinct ESP-matched pairs. For example:

- **DOL as base solvent:** DOL (SCE = 1.41, cyclic ether) has $\mathrm{ESP_{min}} \approx -1.69$ eV. A co-solvent with $\mathrm{ESP_{min}} \approx -1.69$ eV from a structurally distinct class would serve as the DOL ESP-matched partner. TMP (-1.74 eV) is too strong a donor by ~ 0.05 eV — outside the matching threshold. The DOL-matched partner is a weaker phosphate ester or a sulfoxide analogue. This constitutes an open derivation.
- **2-MeTHF as base solvent:** 2-MeTHF (SCE = 1.55) sits above SCE* in the dataset. ESP-matched co-solvents for 2-MeTHF would *not* be designed to raise SCE further; they would instead be used in the context of the anion receptor mechanism (Preprint 3) to bring SCE down from above.

7.3 Extension to Other Metal Ion Chemistries

The ESP-matching principle is not Li-specific. For Na^+, K^+, Zn^{2+}, or Al^{3+} batteries, the same procedure applies: identify SCE* for the target ion (each has its own fixed point, derivable from the same calculus applied to a metal-ion-specific dataset), then apply the ESP-matching rule to identify co-solvents that drive SCE toward that target. The independent replication by Luo *et al.* (Joule, 2025) [7] confirming that solvation configurational entropy governs performance across four metal ion chemistries establishes that the variable is universal. The design mechanism inherits that universality.

8 Comparison with Prior Co-Solvent Selection Approaches

Table 3: Comparison of co-solvent selection criteria. ESP-matching is the only criterion that selects by electrostatic potential minimum at the coordinating site for the explicit purpose of population splitting and coordination entropy tuning.

Criterion	What it selects	What it misses	Reference basis
Donor number (DN)	Average binding enthalpy to Lewis acid reference	ESP surface topology; geometric distinction of binding site	[4]
Oxidative stability	Solvents stable at cathode potential	No solvation geometry information	[2]
Viscosity / conductivity	Fluidity and ionic transport	No solvation structure information	[6]
Film-forming additives	SEI decomposition chemistry	Primary solvation shell geometry	[1]
ESP matching (this work)	Electrostatically indistinguishable co-solvents from structurally distinct classes	*(none identified; fully novel criterion)*	This work

The ESP-matching criterion occupies a gap in the prior design space. It is not a refinement of existing criteria; it is an orthogonal selection axis that the prior literature does not contain.

9 Discussion

9.1 What the Mechanism Adds to the Framework

Preprint 1 established the target: $\mathrm{SCE}^* = 1.466$. The present paper establishes the first rationally designed mechanism for reaching it from below. The contribution is not just a candidate formulation; it is a *design principle* — a rule that converts the abstract navigation instruction "add structural diversity to the shell" into a concrete, computable, and experimentally actionable criterion: find a co-solvent from a structurally distinct class whose $\mathrm{ESP}_{\mathrm{min}}$ matches your base solvent within $k_B T$.

This principle is computable before synthesis. $\mathrm{ESP}_{\mathrm{min}}$ values can be calculated in hours from standard quantum chemistry packages. The screening criterion filters the chemical universe to a small set of candidates without requiring a single experiment. The experiment confirms or falsifies; it does not discover.

9.2 Why the Preprint Order Matters

Candidate 4 (DME/TMP) is listed as Preprint 2 in the publication plan rather than Preprint 3 (DOL/TMB) because the ESP-matching mechanism is the simpler and more broadly applicable of the two mechanisms introduced in the series. The anion receptor mechanism (Preprint 3) requires an additional layer of reasoning (the geometric correction for approach from above SCE^*). Establishing the ESP-matching principle first gives the reader the conceptual foundation for understanding why the Preprint 3 direction choice (DOL rather than DME as the base solvent) is a necessary consequence of the framework, not an arbitrary selection.

9.3 The Predicted Performance and Its Interpretation

The predicted $\mathrm{CE}_{\mathrm{RT}}^{\mathrm{pred}} = 73\%$ at SCE^* may appear low relative to the best published RT performance values in the dataset ($> 99\%$ for HCE/LHCE systems). This requires interpretation.

The 73% prediction is the Regime 1 equation evaluated at SCE^*: it represents the *geometry-driven component* of $\mathrm{CE}_{\mathrm{RT}}$ in the absence of concentration effects, film-forming additives, or other optimisations. It is not the ceiling for a fully optimised DME/TMP formulation; it is the floor set by the solvation geometry alone.

The framework predicts that as the system is optimised (salt concentration, additive package, electrode pre-treatment), $\mathrm{CE}_{\mathrm{RT}}$ will rise above 73% while $\mathrm{CE}_{\mathrm{LT}}$ simultaneously rises above 61% — because the underlying solvation geometry is at the correct fixed point. The key experimental test is not the absolute CE values but the *simultaneous improvement of both* as the system is navigated toward SCE^*.

9.4 The Regime Boundary Risk

The primary risk in the DME/TMP ratio scan is inadvertent regime crossing. If LiFSI salt concentration is increased (to improve $\mathrm{CE}_{\mathrm{RT}}$), the system may enter Regime 2, where the RT–SCE gradient inverts. Adding TMP in a Regime 2 system would raise SCE *away* from the optimal direction for a Regime 2 system, decreasing $\mathrm{CE}_{\mathrm{RT}}$ rather than increasing it.

The protocol specifies 1.0–1.2 M LiFSI to remain in Regime 1. If experiments are conducted at higher concentrations, the three-regime classification must be checked first, and the applicable equations applied. This is not a failure of the framework; it is a consequence of the regime structure that the framework itself predicts and maps.

9.5 Limitations

- The ESP values in Table 1 are computed at a specific level of theory and implicit solvent model. Gas-phase ESP values differ from solution-phase values. The matching condition ($|\Delta\text{ESP}| \leq k_B T$) should be verified at the level of theory appropriate for the solvent environment of interest.

- The Raman-based SCE estimation protocol is approximate. Raman peak shifts provide qualitative confirmation of coordination changes but not precise population fractions. MD simulation is the preferred method for quantitative SCE determination at the target ratio.

- The predicted CE values use Regime 1 equations fit to a dataset of 8 systems. The confidence interval on the predictions is correspondingly wide. The directional prediction (both CEs improve) is robust; the absolute numerical predictions are estimates.

- TMP is known to be flammable and has moderate toxicity. Practical formulation for commercial applications will require fluorinated or otherwise stabilised phosphate ester variants (e.g., Candidate 4b class).

10 Conclusion

We have introduced the **ESP-matching mechanism** as the first rationally designed principle for approaching the Li metal battery performance fixed point $\text{SCE}^* = 1.466$ from below.

The mechanism operates as follows: when a co-solvent is selected from a structurally distinct chemical class whose ESP minimum at the Li^+ coordinating site matches that of the primary solvent within thermal energy ($|\Delta\text{ESP}| \leq k_B T$), Li^+ cannot electrostatically prefer either species. Both coordination geometries are simultaneously populated. The dominant configuration fraction falls. SCE rises toward SCE^*.

DME and TMP are the canonical realisation of this principle. Their ESP minima differ by 0.007 eV — 27% of $k_B T$ at 298 K. They belong to structurally distinct chemical classes (linear ether vs. phosphate ester). Their coordinating atoms, geometries, and Raman signatures are distinguishable. TMP is confirmed as the universal cross-class attractor from all nine ether-anchored molecular searches in the accessible commercial solvent space.

Two candidate formulations are pre-registered with exact falsification conditions: Candidate 4 (LiFSI 1.0–1.2 M in DME/TMP, ratio scan) and Candidate 4b (LiFSI in DME/fluorophosphonate, donor strength comparison). Both are fully specified and testable.

No prior work selects co-solvents by ESP matching to the primary solvent for the purpose of Li^+ coordination entropy tuning. The principle is new. The mechanism is computable before synthesis. The experimental test is bounded and falsifiable.

Data and derivation record availability. All pre-registered predictions, falsification conditions, ESP calculations, molecular search records, and derivations referenced in this paper are archived with full timestamped commit history at:
https://github.com/Eric-Robert-Lawson/attractor-oncology
The repository commit history constitutes the priority and pre-registration record for all claims in this paper.
Relationship to other papers in this series. This is Preprint 2 of the SCE Framework series. Preprint 1 (foundation, fixed point derivation) is available at the same repository. Preprint 3 (anion receptor mechanism, DOL/TMB) is in preparation.

Competing interests. The author declares no competing financial interests.
Author contributions. E.R.L.: conceptualisation, derivation, ESP analysis, writing.

A ESP Calculation Protocol

Electrostatic potential surface minima were computed as follows:

1. Geometry optimisation at M06-2X/6-31G(d) in the gas phase.
2. Single-point energy and ESP calculation at M06-2X/6-311+G(d,p) with the SMD implicit solvent model ($\epsilon = 7.2$, representative of the ether solvent environment).
3. ESP_{min} identified as the most negative value on the 0.001 a.u. electron density isosurface at the primary coordinating atom.
4. For DME: ESP_{min} at the ether oxygen in the gauche conformation (relevant for Li^+ bidentate coordination).
5. For TMP: ESP_{min} at the phosphoryl oxygen (P=O group, primary Li^+ coordinating site).

Full input files and output logs are available in the repository under `Principles_First_Full_Derivation/Const`
`ESP_Calculations/`.

B TMP Physical Properties Summary

Table 4: Physical properties of TMP relevant to electrolyte formulation.

Property	Value	Source
Melting point	−46 °C	CRC Handbook [3]
Boiling point	197 °C	CRC Handbook [3]
Density (25 °C)	1.214 g mL^{-1}	CRC Handbook [3]
Dielectric constant (25 °C)	20.6	[10]
Donor number	23	[4]
ESP_{min} (this work)	−1.7402 eV	This work
Electrochemical window (Li metal)	Stable at 0–4.5 V vs. Li/Li^+	[12]
Availability	Commercial (Sigma-Aldrich, TCI, etc.)	—

C Pre-Registered Falsification Conditions (Verbatim)

The following conditions are reproduced verbatim from the repository commit record. The commit timestamp predates this document.

Candidate 4 (LiFSI 1.0–1.2M, DME/TMP ratio scan):

- *Primary falsification:* Framework falsified if CE_{RT} and CE_{LT} do not both improve as TMP fraction increases from 0 toward the identified target ratio.

- *Secondary falsification:* Framework falsified if at the ratio confirmed by Raman to have SCE ≈ 1.466: $CE_{RT} < 84\%$ OR $CE_{LT} < 60\%$.

- *Mechanistic falsification:* ESP-matching mechanism falsified if Raman shows no P=O shift on TMP addition (TMP not entering Li^+ primary shell).

Candidate 4b (LiFSI 1.0M, DME/fluorophosphonate):

- *Mechanism falsification:* Falsified if DMFP and TMP produce identical SCE at identical volume ratios (no donor strength dependence confirmed).

- *Performance falsification:* Falsified if CE_{RT} and CE_{LT} do not both improve as DMFP fraction increases toward the identified target ratio.

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